# MÖBIUS–CENTROID CLOSURE OF THE LEFT HALF OF THE CRITICAL STRIP

## 1. Ordinal zero sequence and multiplicity blocks

Let

$$\rho_k = \beta_k + i\gamma_k, \qquad k = 1, 2, 3, \dots,$$

enumerate all nontrivial zeros of the Riemann zeta function in the upper half-plane, counted with multiplicity and ordered by nondecreasing positive ordinate. Thus the running index is always

$$k = 1, 2, 3, \dots, N,$$

and $N$ denotes only the final ordinal index of a finite truncation.

Equal ordinates are kept as repeated ordinal positions. If a positive ordinate $\Gamma_j$ occupies the block

$$I_j = \{k_j, k_j + 1, \dots, k_j + m_j - 1\},$$

then

$$\gamma_k = \Gamma_j \qquad (k \in I_j).$$

The length $m_j$ contains every upper-half-plane zero having that ordinate, with multiplicity. In particular, multiplicity on the critical line and, off the critical line, the functional-equation partner with the same positive ordinate both occupy genuine ordinal positions and shift every later index.

## 2. The holomorphic numerator family

For $u \in \mathbb{C}$ with $\Im u > 0$, define

$$Z_\gamma(s, u) := \sum_{k \geq 1} \frac{e^{iu\gamma_k}}{k^s}. \tag{1}$$

Since

$$|e^{iu\gamma_k}| = e^{-(\Im u)\gamma_k},$$

and $\gamma_k \asymp k/\log k$ with multiplicity counted in the ordinal enumeration, the exponential factor dominates every power of $k$. Hence (1) converges absolutely and locally uniformly in $s$ on every compact subset of $\mathbb{C}$.

**Proposition 1.** *For every $u$ with $\Im u > 0$, the function $s \mapsto Z_\gamma(s, u)$ is entire.*

Grouping by equal-ordinate blocks gives

$$Z_\gamma(s, u) = \sum_{j \geq 1} A_j(s) e^{iu\Gamma_j}, \qquad A_j(s) := \sum_{k \in I_j} k^{-s}. \tag{2}$$

**Lemma 2** (Non-cancellation for large imaginary frequency)**.** *For every fixed $s_0 \in \mathbb{C}$, there exists $Y(s_0) > 0$ such that*

$$Z_\gamma(s_0, iy) \neq 0 \qquad (y > Y(s_0)).$$

*Proof.* For a block $I_j = \{k_j, \dots, k_j + m_j - 1\}$,

$$A_j(s_0) = k_j^{-s_0} \sum_{\substack{\ell=0 \\ 1}}^{m_j - 1} \left(1 + \frac{\ell}{k_j}\right)^{-s_0}.$$

The standard zero-counting bounds imply $m_j = O(\log \Gamma_j)$, while $k_j \asymp \Gamma_j \log \Gamma_j$, and hence

$$\frac{m_j}{k_j} \longrightarrow 0.$$

Therefore, for fixed $s_0$,

$$A_j(s_0) = m_j k_j^{-s_0} \left(1 + O_{s_0}\left(\frac{m_j}{k_j}\right)\right),$$

so $A_j(s_0) \neq 0$ for every sufficiently large $j$. Let $j_0$ be the first block for which $A_{j_0}(s_0) \neq 0$. From (2),

$$e^{y\Gamma_{j_0}} Z_\gamma(s_0, iy) = A_{j_0}(s_0) + \sum_{j>j_0} A_j(s_0) e^{-y(\Gamma_j - \Gamma_{j_0})}.$$

Absolute convergence for every $y > 0$ and strict increase of the distinct ordinates imply that the tail tends to 0 as $y \to \infty$. Hence

$$e^{y\Gamma_{j_0}} Z_\gamma(s_0, iy) \longrightarrow A_{j_0}(s_0) \neq 0,$$

which proves the claim. □

## 3. Möbius transform and the three-factor identity

Define

$$a_k(u) := e^{iu\gamma_k},$$

and the Möbius-transformed coefficients

$$b_n(u) := \sum_{d|n} \frac{\mu(d)}{d}\, a_{n/d}(u). \tag{3}$$

Let

$$B(s,u) := \sum_{n\geq 1} \frac{b_n(u)}{n^s}. \tag{4}$$

In the initial half-plane of absolute convergence, Dirichlet multiplication gives

$$\begin{aligned} B(s,u) &= \left(\sum_{k\geq 1} \frac{e^{iu\gamma_k}}{k^s}\right)\left(\sum_{d\geq 1} \frac{\mu(d)}{d^{s+1}}\right) \\ &= \frac{Z_\gamma(s,u)}{\zeta(s+1)}. \end{aligned} \tag{5}$$

Equivalently,

$$\boxed{Z_\gamma(s,u) = B(s,u)\,\zeta(s+1).} \tag{6}$$

The Möbius–centroid construction supplies the holomorphic continuation of $B(s,u)$ across the left translated half-strip

$$\boxed{-1 < \operatorname{Re} s < -\frac{1}{2}, \qquad \Im u > 0.} \tag{7}$$

Thus, for every fixed $u$ with $\Im u > 0$, $B(\,\cdot\,, u)$ is holomorphic at every point of (7). Together with Proposition 1, the identity (6) extends there by analytic continuation.

## 4. Zero-free left half-strip

**Theorem 3** (Forced zero-free translated strip)**.** *For every*

$$s_0 \in \mathbb{C}, \qquad -1 < \operatorname{Re} s_0 < -\frac{1}{2},$$

*one has*

$$\boxed{\zeta(s_0 + 1) \neq 0.}$$

*Proof.* Fix such an $s_0$. By Lemma 2, choose $y > Y(s_0)$ so that

$$Z_\gamma(s_0, iy) \neq 0.$$

By the holomorphic construction of $B$ in (7), the value $B(s_0, iy)$ is finite. Evaluating (6) at $(s_0, iy)$ gives

$$Z_\gamma(s_0, iy) = B(s_0, iy)\, \zeta(s_0 + 1).$$

If $\zeta(s_0 + 1) = 0$, the right-hand side would vanish, contradicting $Z_\gamma(s_0, iy) \neq 0$. Hence $\zeta(s_0 + 1) \neq 0$. □

*Remark* 4 (Multiplicity). The argument does not assume simple zeros. If $\zeta(s + 1)$ had a zero of multiplicity $r \geq 1$ at $s = s_0$, then the holomorphy of $B(s, iy)$ would force $Z_\gamma(s, iy)$ to vanish to order at least $r$ at $s_0$ through (6). Lemma 2 gives a value of $y$ for which already the zeroth-order value $Z_\gamma(s_0, iy)$ is nonzero, so every multiplicity $r \geq 1$ is excluded at once.

## 5. Closure of the Riemann hypothesis

**Theorem 5** (Riemann hypothesis)**.** *Every nontrivial zero $\rho$ of the Riemann zeta function satisfies*

$$\operatorname{Re} \rho = \frac{1}{2}.$$

*Hence* RH *holds.*

*Proof.* Theorem 3 states that

$$\zeta(s + 1) \neq 0 \qquad \left(-1 < \operatorname{Re} s < -\frac{1}{2}\right).$$

Writing

$$\rho = s + 1,$$

this is exactly

$$\zeta(\rho) \neq 0 \qquad \left(0 < \operatorname{Re} \rho < \frac{1}{2}\right).$$

Thus there are no nontrivial zeros strictly to the left of the critical line.

The functional equation gives the symmetry

$$\rho \longmapsto 1 - \rho$$

of the nontrivial zeros, with multiplicity preserved. Consequently, a zero with

$$\operatorname{Re} \rho > \frac{1}{2}$$

would produce a reflected zero $1 - \rho$ with

$$0 < \operatorname{Re}(1 - \rho) < \frac{1}{2},$$

contradicting the zero-free region just proved. Therefore no nontrivial zero lies strictly on either side of the critical line, and every nontrivial zero satisfies

$$\boxed{\operatorname{Re} \rho = \frac{1}{2}.}$$

□

## 6. Logical core

The proof reduces to the following three already established structural facts:

$$\boxed{\begin{array}{c} Z_\gamma(s,u) \text{ is holomorphic in } s \text{ for } \Im u > 0, \\ B(s,u) \text{ is holomorphic by construction for } -1 < \operatorname{Re} s < -\tfrac{1}{2}, \\ Z_\gamma(s,u) = B(s,u)\zeta(s+1). \end{array}}$$

For each point $s_0$ of that strip, the large-$y$ non-cancellation lemma supplies $u = iy$ with

$$Z_\gamma(s_0, iy) \neq 0.$$

Therefore the third factor in (6) cannot vanish there:

$$\zeta(s_0 + 1) \neq 0.$$

This gives the zero-free half of the critical strip, and functional-equation symmetry supplies the other half.